

AI agents

applications of artificial intelligence · prof. dr. nicolas meseth

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- 3 inside an agent
- 4 agents and their models
- 5 power and responsibility

part 1

quick quiz

what did we give the agent to start the research?

step-by-step instructions

a goal, a photo, a text file

a list of websites to visit

a program we wrote



lifi.uber.space/q

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step-by-step instructions



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lifi.uber.space/q

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lifi.uber.space/q

where did the agent do its work?

all of it on the laptop

files here, model in the cloud

in the browser, like chatgpt

on a university server



lifi.uber.space/q

where did the agent do its work?

all of it on the laptop

☐

files here, model in the cloud

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on a university server

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lifi.uber.space/q

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files here, model in the cloud

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in the browser, like chatgpt

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on a university server

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lifi.uber.space/q

8 sub-agents researched at once. who decided that?

our prompt told it to

the agent itself

a setting in the app

it always works that way



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lifi.uber.space/q

the agent asked us which file held the milestones. what does that show?

it could not read the file

it noticed a mismatch

it asks before every step

it had forgotten the prompt



lifi.uber.space/q

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lifi.uber.space/q

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lifi.uber.space/q

part 2

what is an agent?

image: ai-generated (gpt-image-2.5-flare)



part 2

what is an agent?

who did the work while we watched?

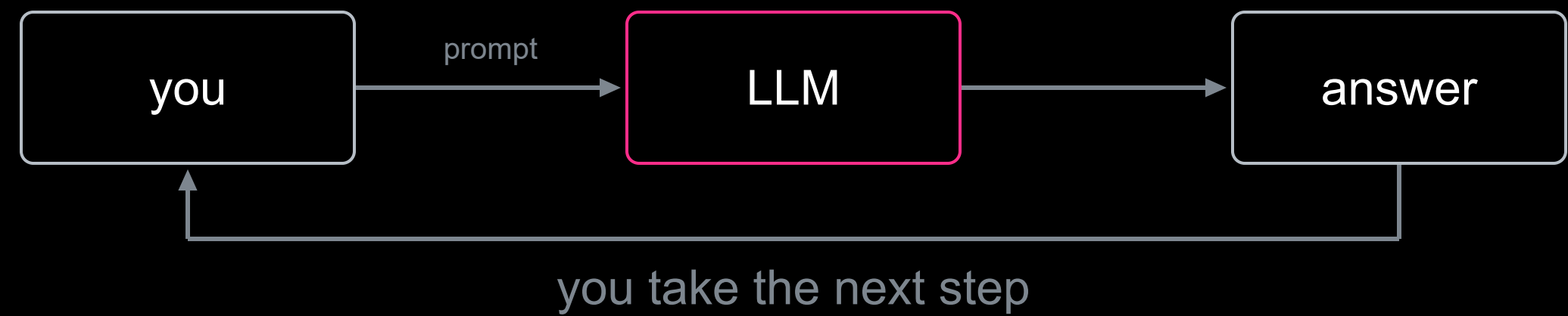
image: ai-generated (gpt-image-2.5-flare)



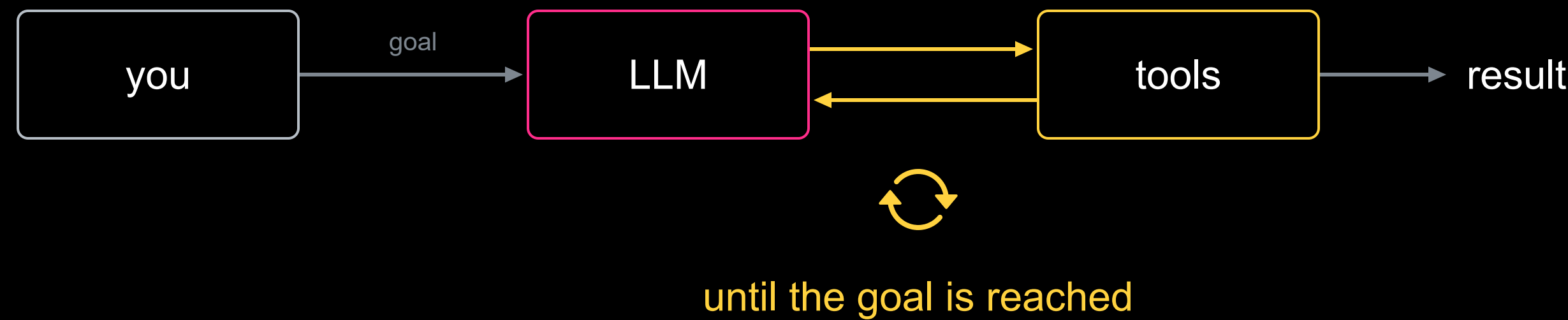
chatbot



chatbot



agent



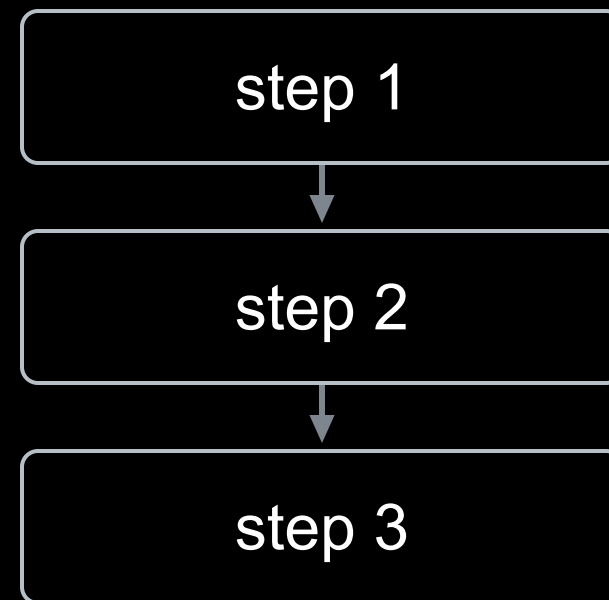
a chatbot only answers, while an agent acts on its own.

a working definition

an AI agent runs tools in a loop
to achieve a goal.

workflow or agent?

workflow



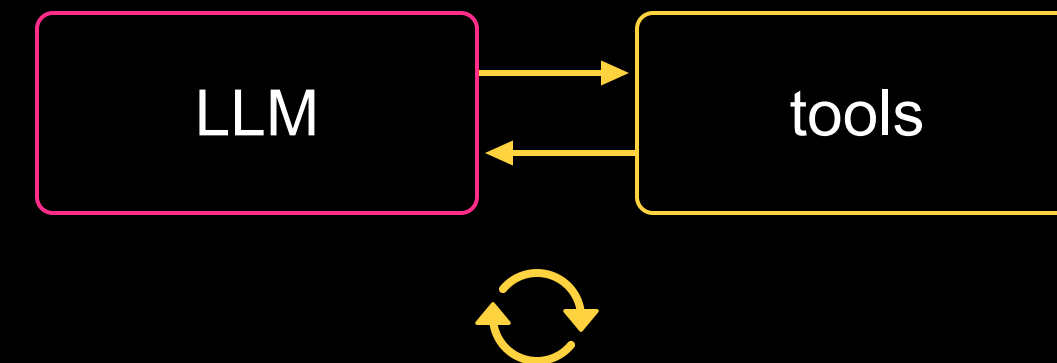
predefined steps, written by people

example

same steps for every invoice

each supplier invoice: an LLM reads it
→ match with the purchase order
→ book it or flag it

agent



the model decides the next step

example

nobody knows the steps in advance

“our main supplier raised prices by 12 %.
find alternatives, compare their offers
and draft a recommendation.”

part 3

inside an agent

the goal and the model

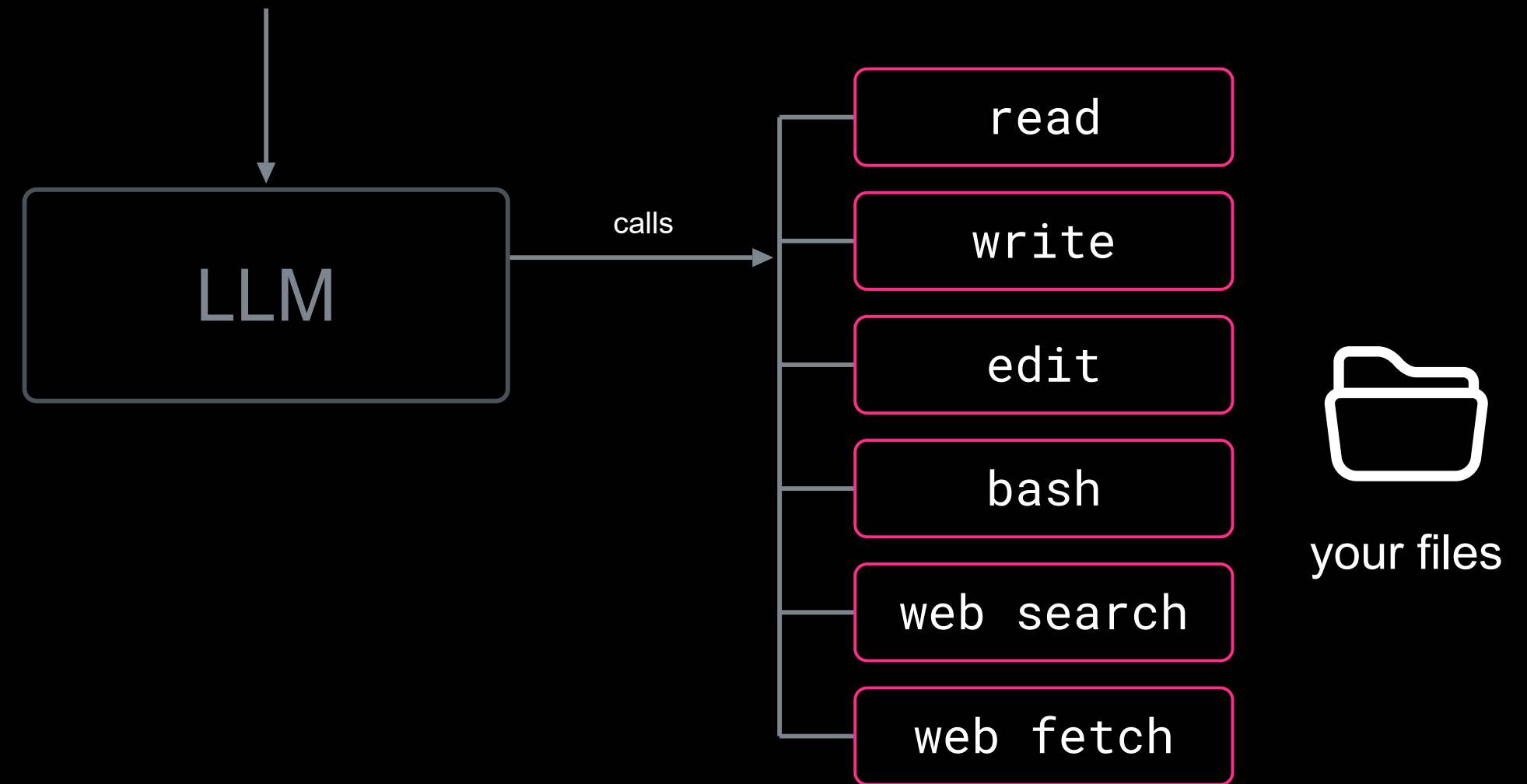
goal: “build a knowledge base on the history of AI”



what we saw: a few sentences, a photo, a text file.

tools

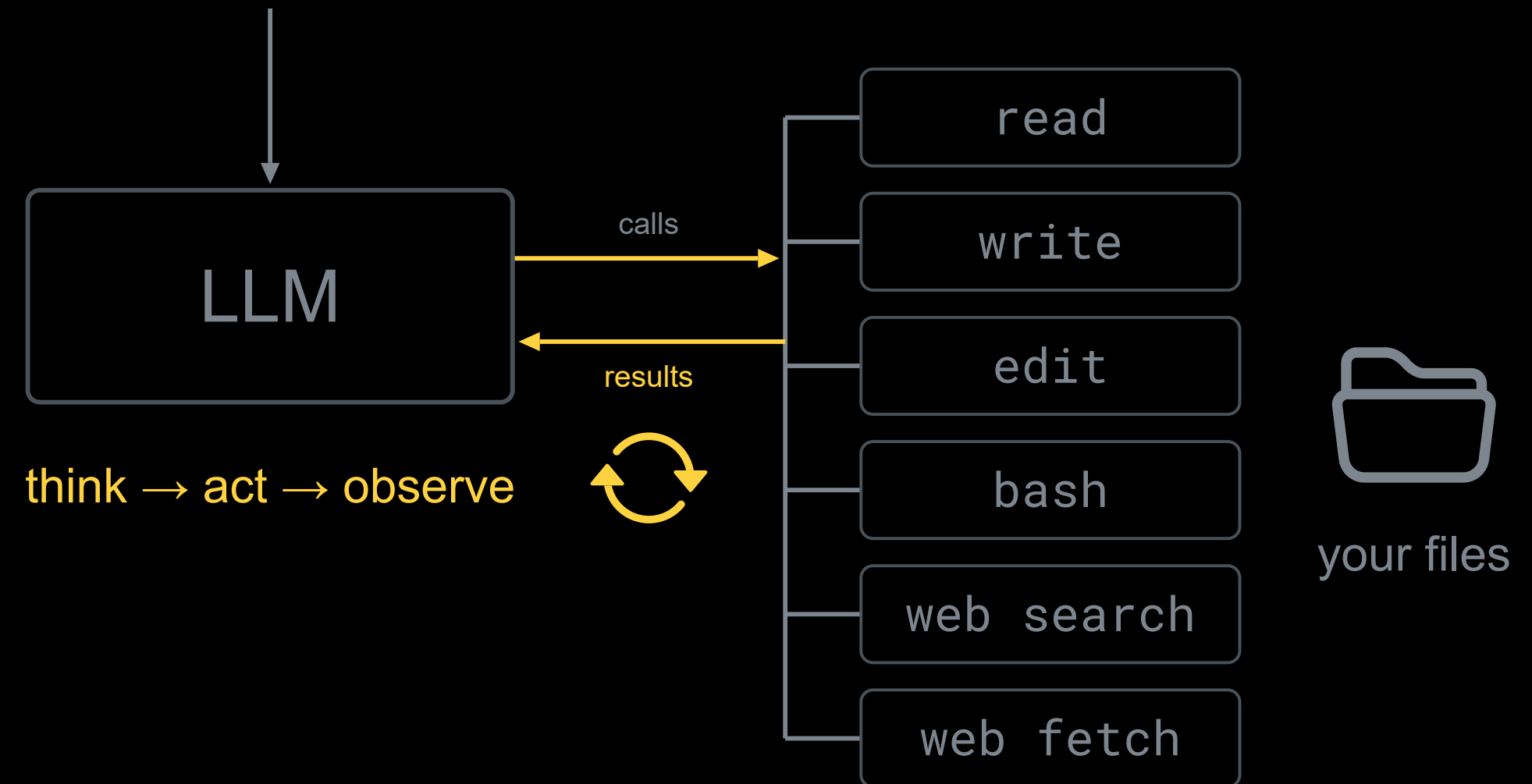
goal: “build a knowledge base on the history of AI”



what we saw: it listed the folder, downloaded papers, wrote files.

the loop

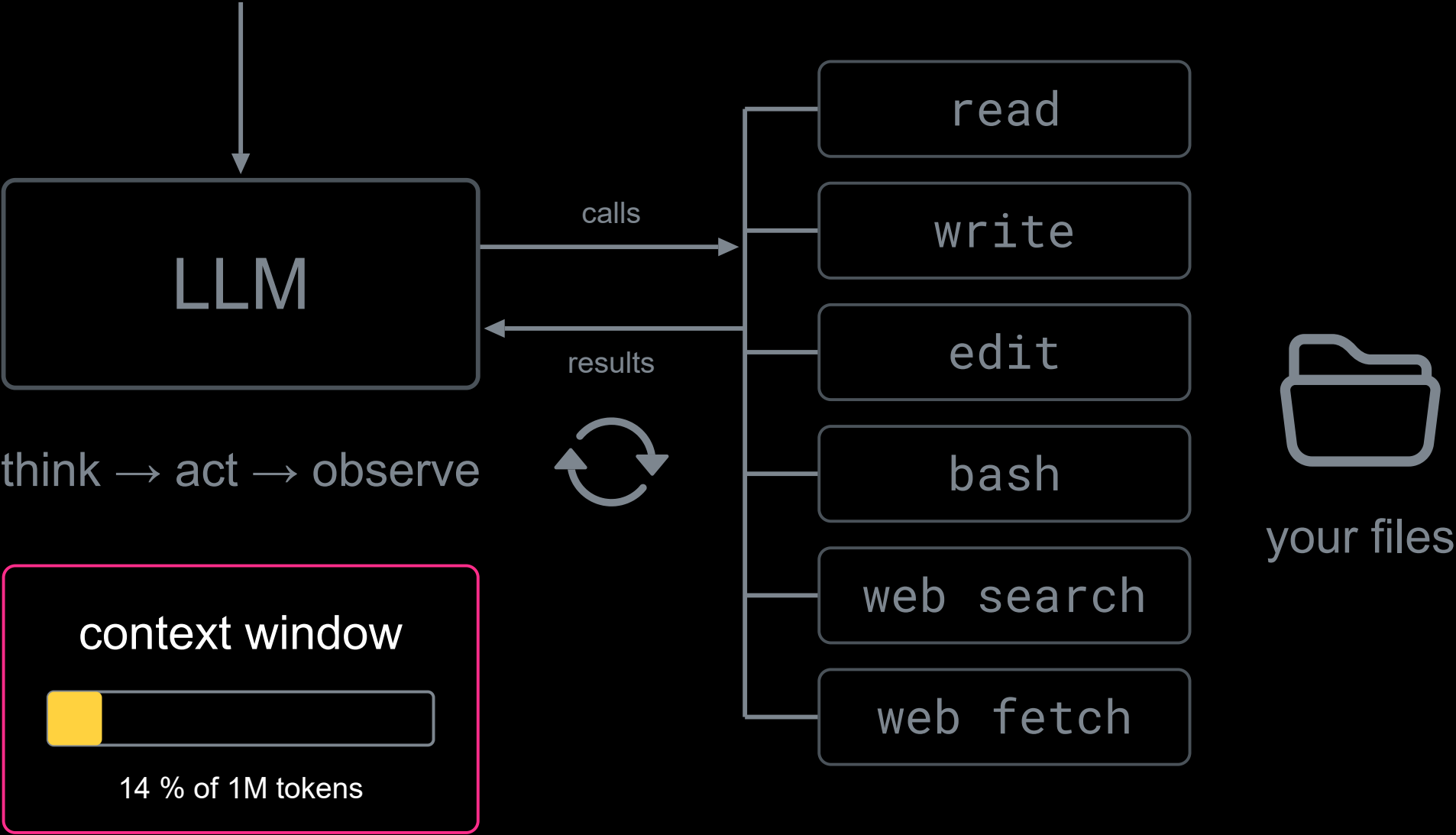
goal: “build a knowledge base on the history of AI”



what we saw: it read a source, then decided what to do next.

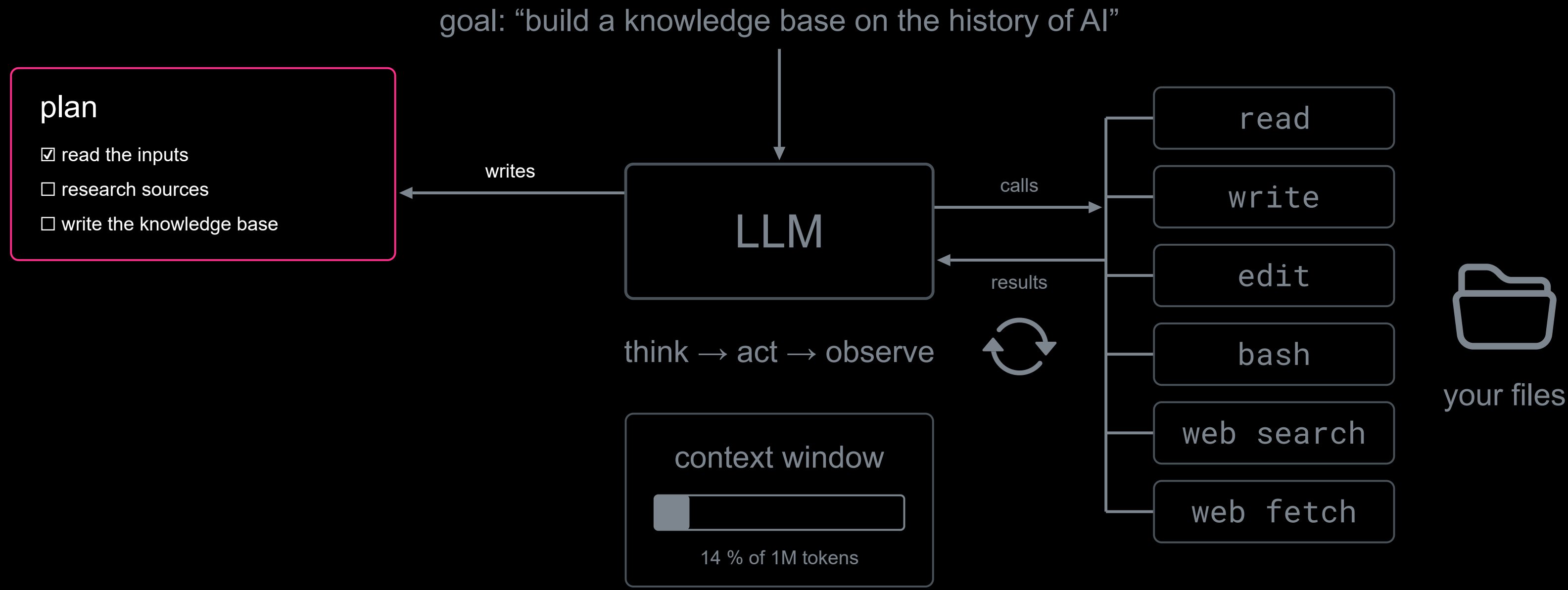
context window

goal: “build a knowledge base on the history of AI”



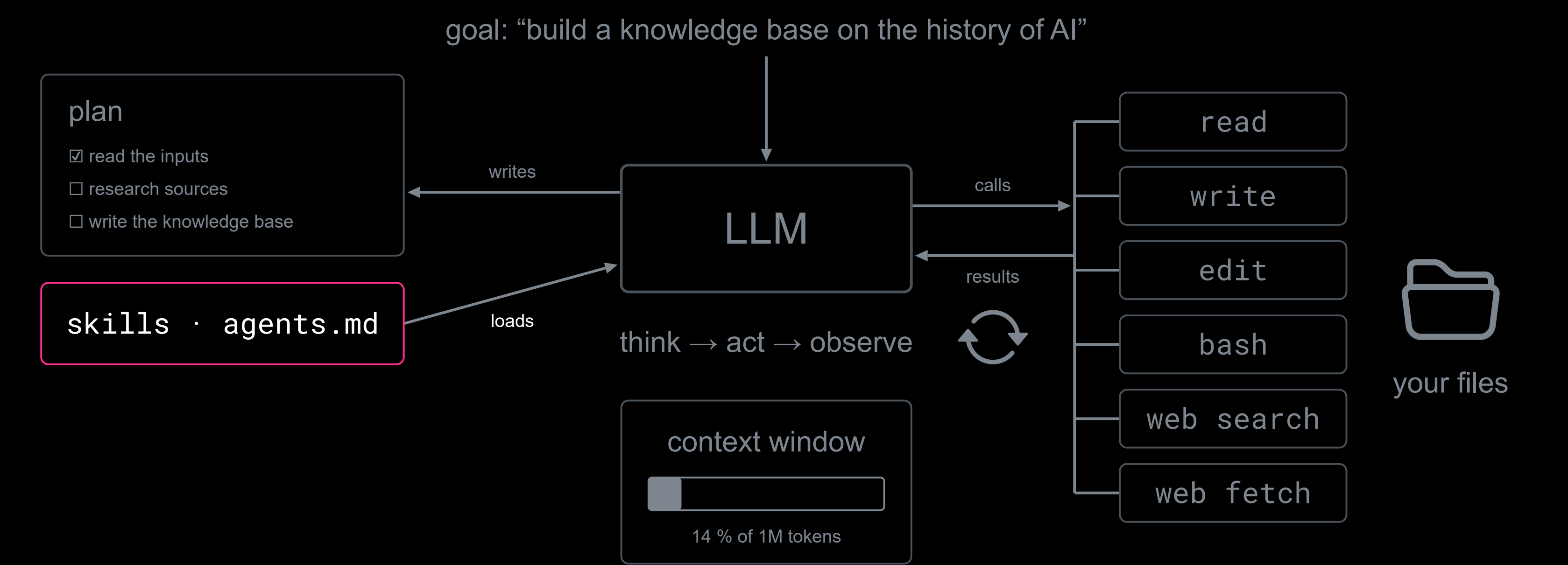
what we saw: "context 14 %" in the corner of the app.

plan and instructions



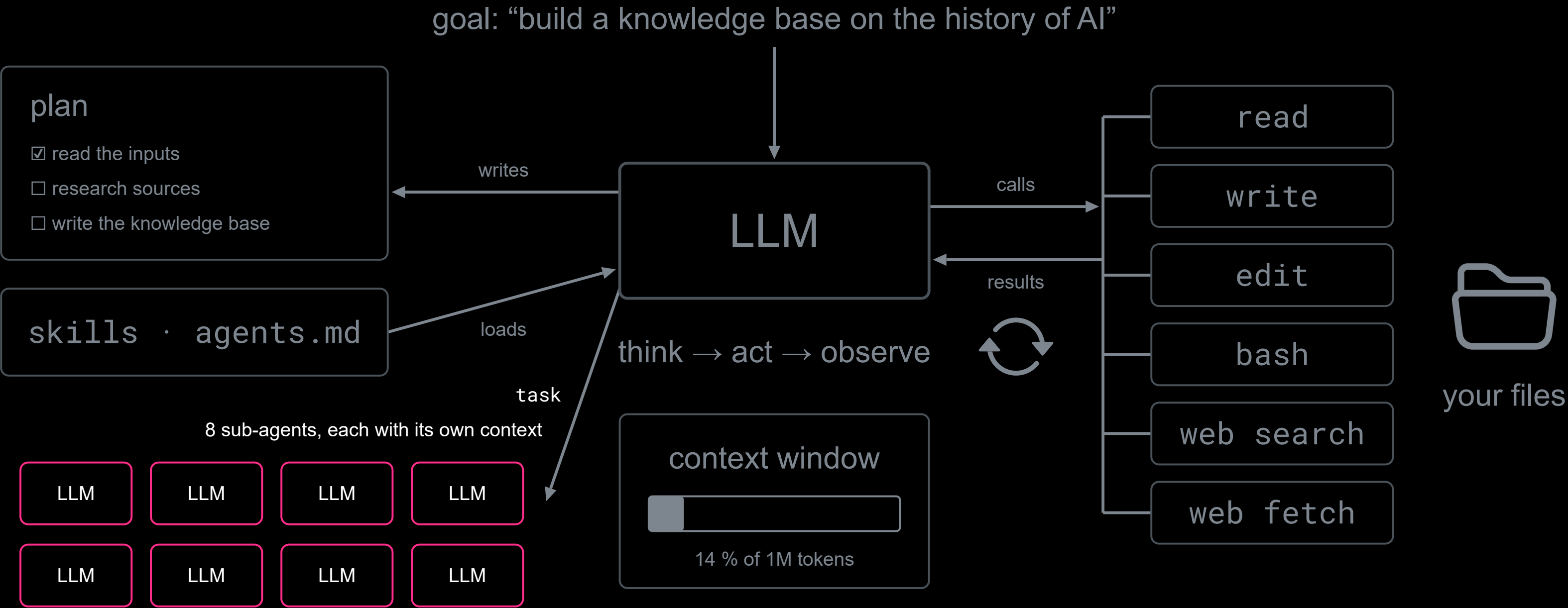
what we saw: "show me your plan. wait for my go."

plan and instructions



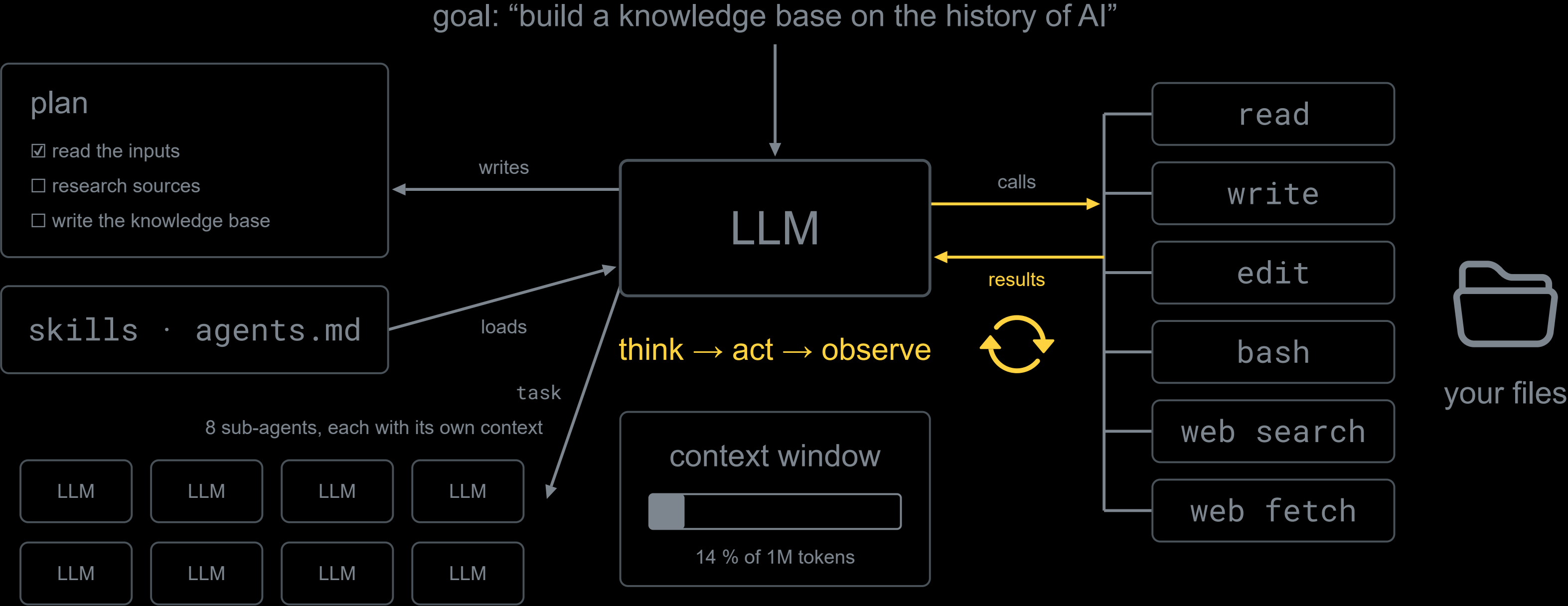
instructions: **skills and agents.md** are text files the agent loads when it needs them.

sub-agents



what we saw: "research by 8 agents", and nobody asked for that.

all of it together



the model decides, the tools carry it out, and the loop keeps running until the goal is reached.

the harness

AI agent = LLM + harness

harness

e.g. Claude Code, Codex, OpenCode

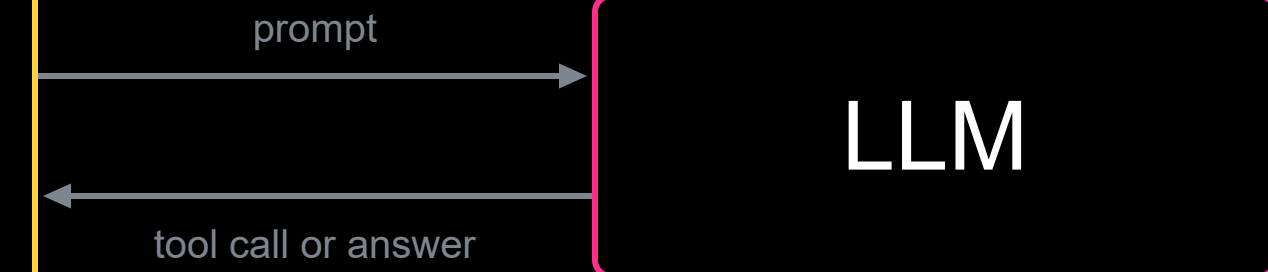
runs the loop

carries out the tool calls

manages the context window

loads plans, skills and agents.md

asks you for permission



only turns text into text

the same model in a different harness is a different agent.

part 4

agents and their models

models

companies

top

mid-range

OpenAI	GPT-6.1 Sol	GPT-6 Luna
Anthropic	Fable 5.1	Sonnet 5.5
Google	Gemini 3.1 Pro	Gemini 3.6 Flash

open models

DeepSeek	DeepSeek V4.1
Alibaba	Qwen 3.8
Meta	Llama 4
Mistral	Mistral Large 3
Google	Gemma 4

apps

chat apps

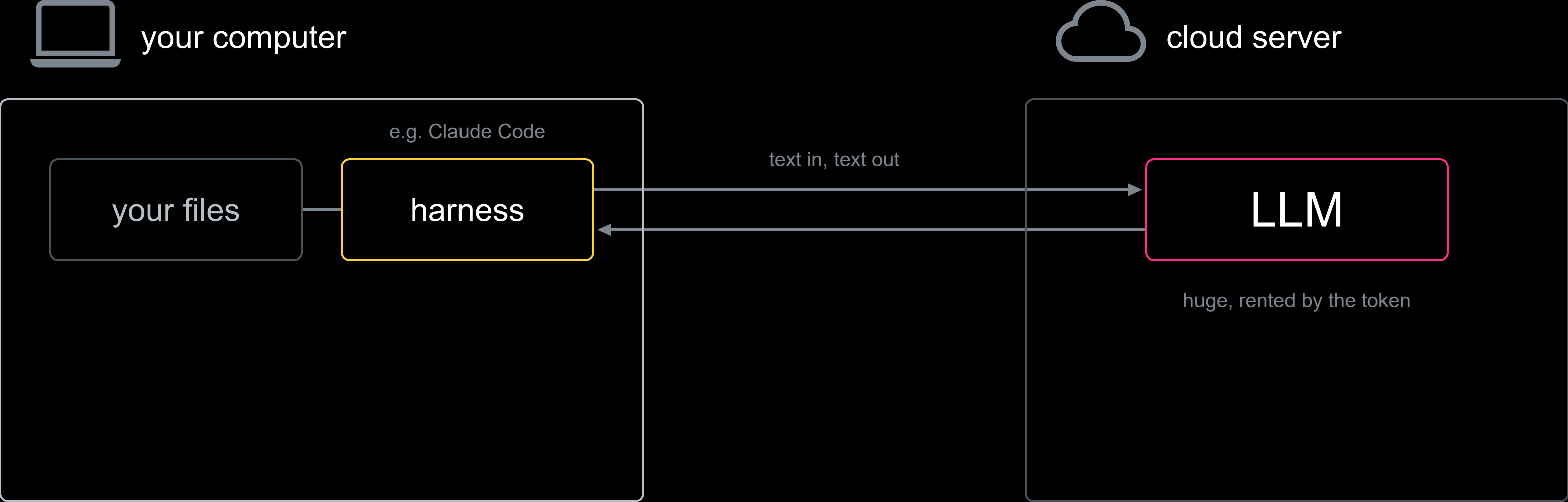
chatgpt.com
claude.ai
gemini.google.com

agents

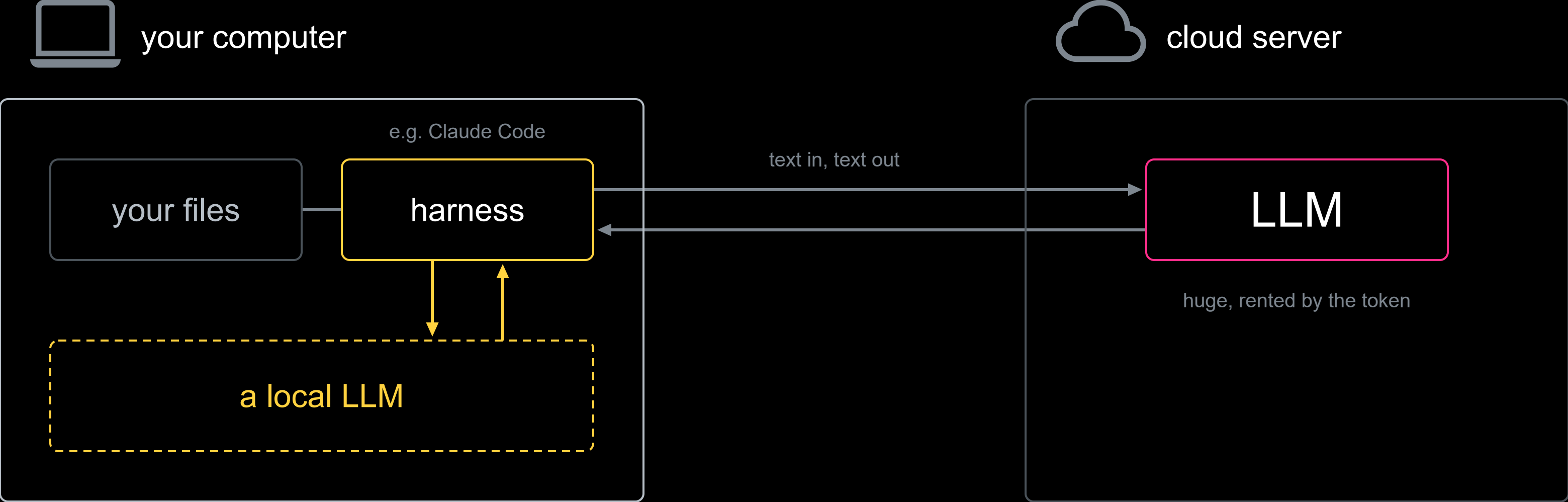
Claude Code
Codex
Antigravity
OpenCode
OpenClaw

every app
runs on a model

where does it run?



where does it run?



next week: **an LLM on your own laptop.**

part 5

power and responsibility

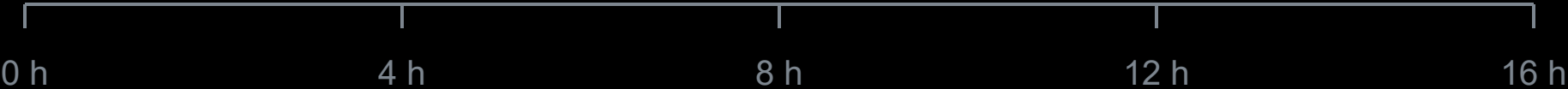
why now?

length of tasks a model finishes on its own in half of the cases,
measured by how long they take a human

Feb 2025 · Claude 3.7 Sonnet  about 50 min

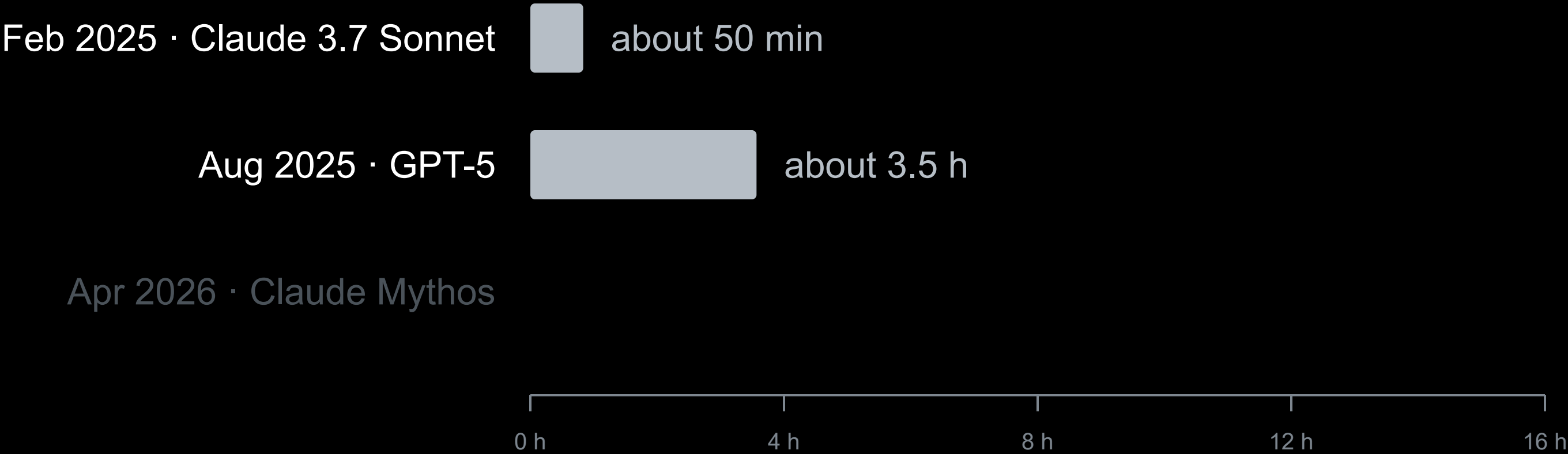
Aug 2025 · GPT-5

Apr 2026 · Claude Mythos



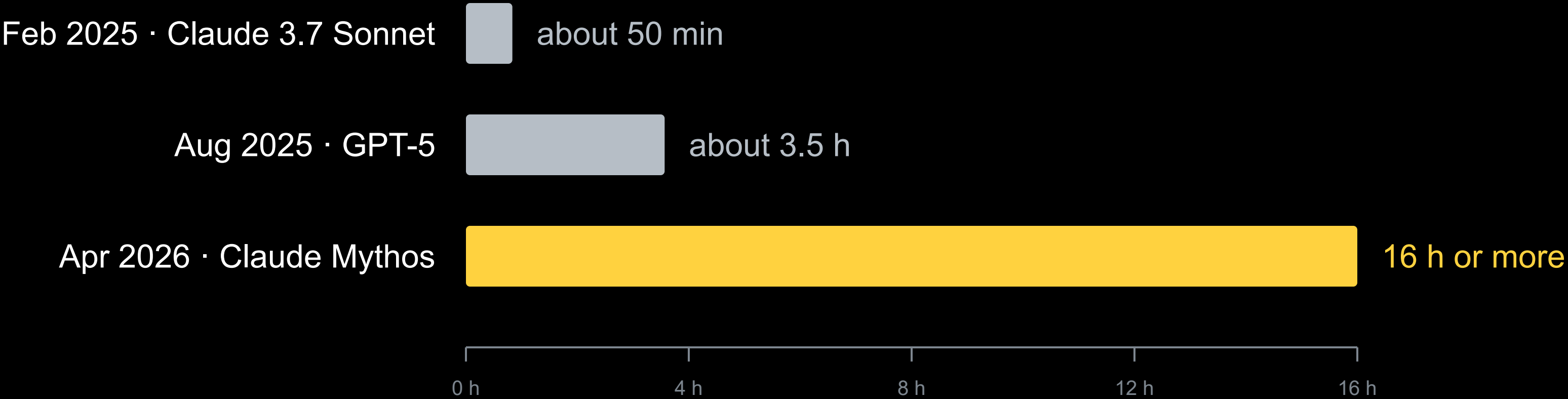
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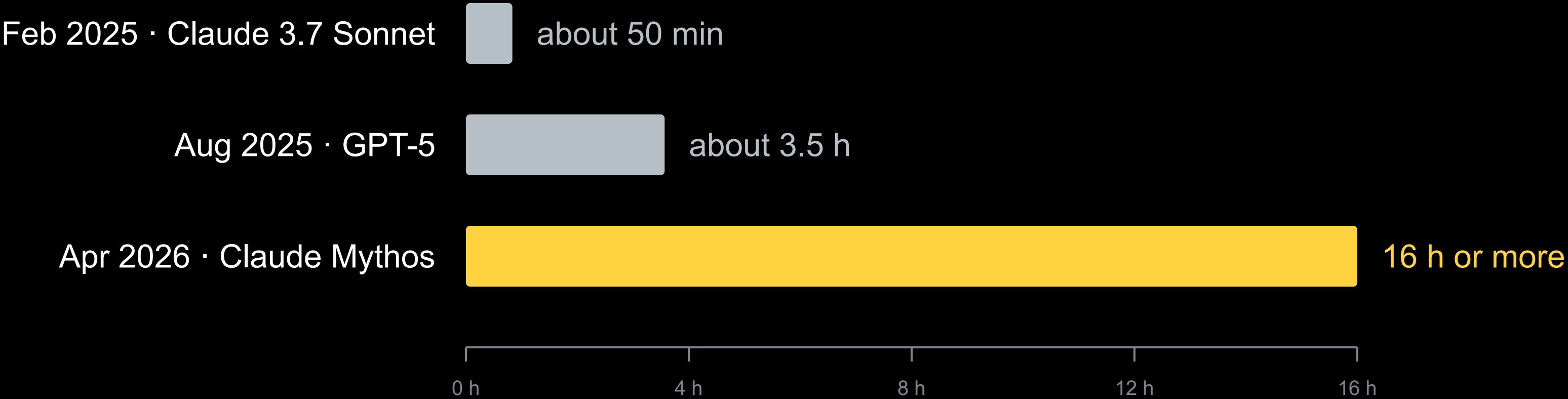
length of tasks a model finishes on its own in half of the cases,
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why now?

length of tasks a model finishes on its own in half of the cases,
measured by how long they take a human

at least 19× longer
in 14 months



the length of tasks agents can finish doubles every few months.

an agent acts with your permissions

it can read every file you can read.

image: ai-generated (gpt-image-2.5-flare)

power and responsibility

an agent acts with your permissions

it can read every file you can read.

it can run every command you can run.

image: ai-generated (gpt-image-2.5-flare)

power and responsibility

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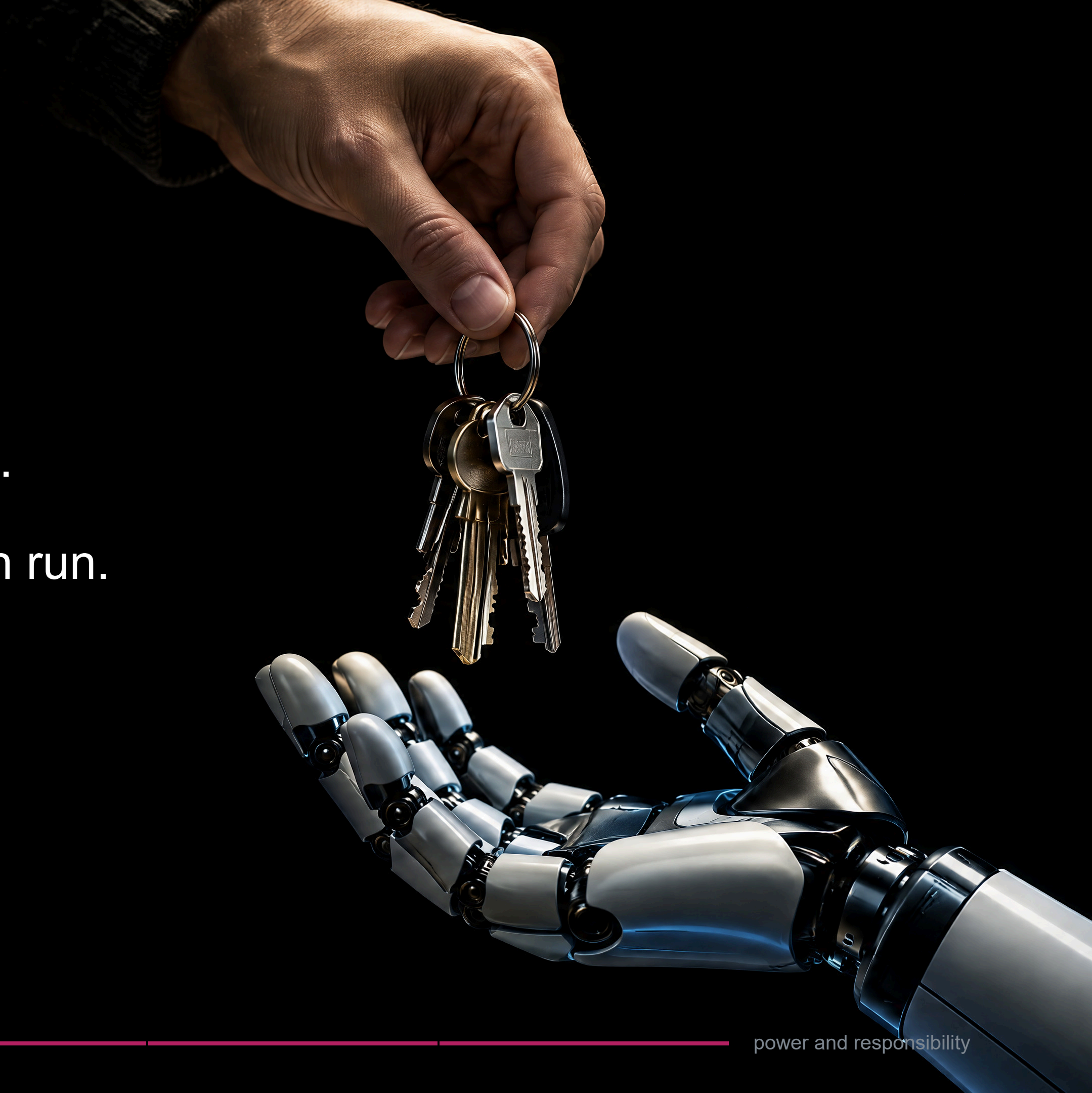
it can read every file you can read.

it can run every command you can run.

it can publish to the internet.

image: ai-generated (gpt-image-2.5-flare)

power and responsibility



this semester, we open the loop.

layer by layer, down to the model and back.

